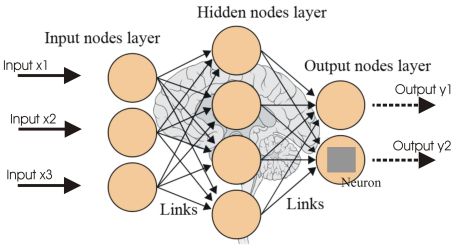
Part 1 - Topics



- **Introduction**
- A little history of neural networks
- The mathematics behind ANN - part I
- The perceptron and the Multi Layer Perceptron
- Neural Networks architectures: an overview

DMB - Machine Learning

1.1 - Introduction

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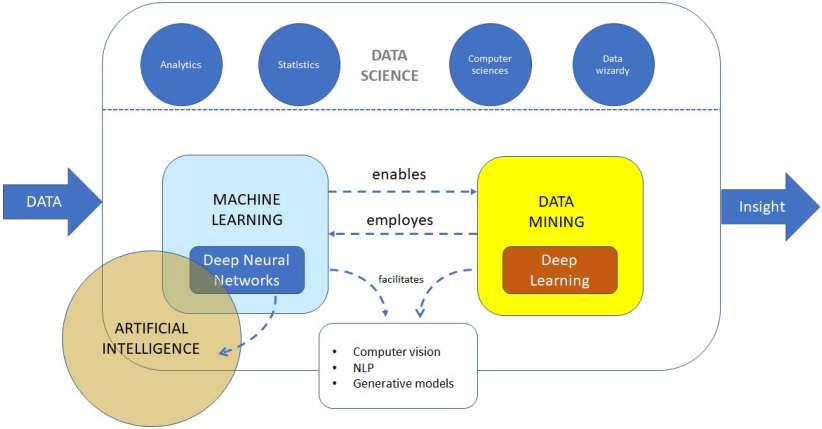
Summary

1. Where are you?
2. What is Data Science?
3. Data and understanding
4. A short recap
5. Tips & resources

Where are you?

You want to become a data scientist, hence it is important to understand where this course places itself in your formation process.



The four pillars of Data Science (©G. Longo).

The four pillars of Data Science

The four blue circles **Analytics**, **Statistics**, **Computer Science(s)** and **Data Wizardy** are the four pillars of data science. You should already have a basic/good understanding of them.

The central part of the diagram is at the core of the two modules of this course. **Machine Learning (ML)** and **Data Mining (DM)** are two different faces of the same medal (with some conceptual more than practical differences).

- DM is designed to extract the rules from large quantities of data. In other words, it covers a collection of methods to determine a particular outcome based on the total of the gathered data.
- DM relies on **human intervention** and is ultimately created for use by people. In other words, without human intervention DM cannot work.
- ML **teaches a computer how to learn and comprehend the given parameters in order to perform complex tasks**. Once fine tuned, **ML does not depend on human influence** or actions. This marks a deep difference with **Classical Programming** (see Fig. 1)

- Machine Learning is a complex field and, as all things which are worth studying it inherits ideas, concepts and methods from many different areas.
- the older and deeper roots of ML methodologies are first of all in **Statistics** and, for what the purpose and design of ANNs are concerned, in **Neurosciences**.
- Nowadays, ML has become a discipline by itself even though further developments are often triggered by advances in Information Technology and Neurosciences.

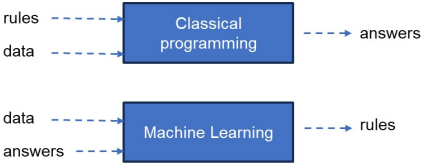
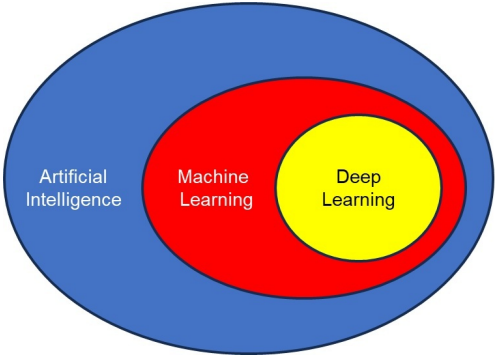


Figure 1: The two different approaches.

ML and AI

... The field of articial intelligence, or AI, seeks to recreate the powerful capabilities of the brain in machines, and today the terms machine learning and AI are often used interchangeably... (Bishop & Bishop, 2024, p.1).

From the barren landscapes inside our personal devices come furtive anthems hummed by those digital servants who will one day be our overlords.... (A.I. Winter)



What is Data Science?

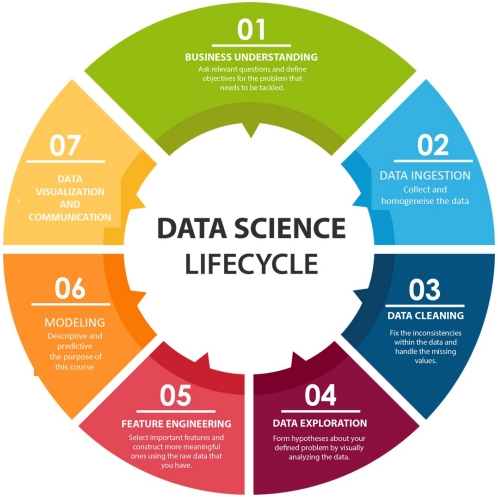
Geometry according to Hilbert

... According to Hilbert, Geometry is what geometrists say it is ...

Definition of what data science is

... Data science is what data scientists say it is ...

So ... let us see what data scientists do... and why the structure of the exam is the way it is ...



Break down of a typical data science project (©adapted by G.Longo).

01 -Business understanding

Before starting a project you must understand the problem. Not from the point of view of the data scientist, but from that of the domain expert who proposed the problem. Hence, you need to aquire a basic understanding of the field in order to proficuously interact with the domain experts. This phase is crucial and usually takes a large fraction of the development phase.

You need to spend quite some time in understanding the properties of the data which you are supposed to use to solve your problem. Data structures are very much domain dependent.

On the contrary of what you might think by reading blogs and on line material**Few things are worse and more dangerous** than jumping from one problem to the other EXPECIALLY AT THE BEGINNING OF YOUR CAREER.

02/03 - Data ingestion and cleaning

This phase is essential for promoting data quality before loading into a data warehouse, data lake, or other repository. In real cases (not toy models) data is (very) expensive and mistakes can cause the demise of the whole project.

The data lifecycle begins with the data collection–both raw **structured** and **unstructured** data from all relevant sources using a variety of methods. These methods can include manual entry, web scraping, and real-time streaming data from systems and devices. Data sources can include structured data, such as customer data, along with unstructured data like log files, video, audio, pictures, the Internet of Things (IoT), social media, and more.

Data management teams help to set standards around data storage and structure, which facilitate workflows around analytics, machine learning and deep learning models. These stages include:

- finding, obtaining, and ingesting the data,
- deduplicating the data, identify obviously wrong data, ecc.
- transforming and combining the data using ETL (extract, transform, load) jobs or other data integration technologies.

04 - Data exploration - EDA

In statistics, exploratory data analysis (EDA) is an approach of analyzing data sets to summarize their main characteristics, often using statistical graphics and other data visualization methods.

A statistical model can be used or not, but primarily EDA is for seeing what the data can tell us beyond the formal modeling and thereby contrasts traditional hypothesis testing.

EDA encompasses initial data analysis (IDA), which focuses more narrowly on checking assumptions required for model fitting and hypothesis testing, and handling missing values and making transformations of variables as needed.

Data scientists conduct an exploratory data analysis to examine biases, patterns, ranges, and distributions of values within the data. This data analytics exploration drives hypothesis generation for a/b testing.

It also allows analysts to determine the datas relevance for use within modeling efforts for predictive analytics, machine learning, and/or deep learning. It includes many different aspects amnong which the most relevant are:

- Data distributions, Understanding the shape of the data is key.
- Handling Missing Data: Address missing values through imputation or removal of instances or features with missing data. There are many algorithmic approaches to handling missing data.

05 - Feature engineering

It includes the so called **Feature Extraction and Selection** i.e. the identification of the most relevant features to improve model interpretability and efficiency using a variety of techniques like univariate feature selection or recursive feature elimination.

It aims at reducing data complexity (often known as data dimensionality) while retaining as much relevant information as possible. This helps to improve the performance and efficiency of machine learning algorithms and simplify the analysis process. Feature extraction may involve the creation of new features (feature engineering) and data manipulation to separate and simplify the use of meaningful features from irrelevant ones. Create new features or reduce dimensionality using techniques such as Principal Component Analysis (PCA) or t-Distributed Stochastic Neighbor Embedding (t-DSNE).

Creating new features gives you a deeper understanding of your data and results in more valuable insights. When done correctly, feature engineering is one of the most valuable techniques of data science, but it is also one of the most challenging.

In practice, **Feature engineering** refers to manipulation addition, deletion, combination, mutation of your data set to improve machine learning model training, leading to better performance and greater accuracy.

An effective feature engineering is based on sound knowledge of the business problem and the available data sources.

Feature engineering is often an iterative process; evaluate, refine, and repeat as needed to achieve optimal results. These steps may vary based on the nature of the data and the machine learning task, but collectively, they contribute to creating a robust and effective feature set for model training

It is worth emphasizing that as you will see in the next lectures, **feature transformation is at the very heart of Machine Learning**.

An example: A common example of feature engineering is when your doctor uses your body mass index (BMI). BMI is calculated from both body weight and height, and serves as a surrogate for a characteristic that is very hard to accurately measure: the proportion of lean body mass.

Feature engineering

- **Variable Encoding**: Convert categorical variables into a numerical format suitable for machine learning algorithms.
- **Feature Scaling**: Standardize or normalize numerical features to ensure they are on a similar scale, improving model performance.
- **Feature Creation**: Generate new features by combining existing ones to capture relationships between variables.
- **Handling Outliers**: Identify and address outliers in the data through techniques like trimming or transforming the data.
- **Normalization**: Normalize features to bring them to a common scale, important for algorithms sensitive to feature magnitudes.
- **Binning or Discretization**: Convert continuous features into discrete bins to capture specific patterns in certain ranges.

Specialised feature analysis

- **Text Data Processing**: If dealing with text data, perform tasks such as tokenization, stemming, and removing stop words.
- **Time Series Features**: Extract relevant timebased features such as lag features or rolling statistics for time series data.
- **Vector Features**: Vector features are commonly used for training in machine learning. In machine learning, data is represented in the form of features, and these features are often organized into vectors.

06 - Modeling

Can be either **Descriptive** or **Predictive**.

- **Descriptive Modeling**. It mostly aims at identifying patterns or relationships in datasets. It serves as an easy way to explore the properties of the data examined earlier and not to predict new properties. In a classification technique, you typically have historical data called labeled examples and new examples.
- **Predictive Modeling**.It aims at predicting future behavior. Predictive modeling solutions are a form of data-mining technology that works by analyzing historical and current data and generating a model to help predict future outcomes.

... It will be covered during this course ...

07 - Data visualization and Communication

This aspects must not be overlooked. At the end, in fact, your success depends on how effective you are in communicating the results of your analysis.

Insights are presented as reports and other data visualizations that make the insightsand their impact on the specific problem easier for analysts and other decision-makers to understand. In this phase it is crucial to remember that the customer, domain scientists, ecc. **know little, if any, about the methodology of data science.**

Lots of care needs to be put to achieve conciseness, clarity and robustness in the description of the results.

Any data science programming language such as Python, MatLab or R, includes components for generating visualizations; alternately, data scientists can use dedicated visualization tools.

Data and understanding

The data

We should define what data is. If you think about it, this is an almost impossible task. As many other words, data has such a broad meaning that you can find hundreds of different definitions, either too restrictive or too broad or incomplete or plainly useless.

- In **Philosophy** data is defined as: things known or assumed as facts, making the basis of reasoning or calculation.
- In **Computer sciences** data is the quantities, characters, or symbols on which operations are performed by a computer, which may be stored and transmitted in the form of electrical signals and recorded on magnetic, optical, or mechanical recording media.
- Or: facts and statistics collected together for reference or analysis ... etc....

Hence: we shall assume the concept of data to be primitive

Why are data so important to understanding? ... and why did they pave the road to ML and therefore to the recent explosion of AI?

Because our representation of the world (our intelligence of the world) is based on data. These data are streamed by our senses in a rich variety.

Some fields of artificial intelligence are trying to imitate the way nature thought us how to extract knowledge from data. **As we shall see, many developments in ML come from an attempt to imitate nature**.

By being exposed to many snapshots of baby bottles, they quickly learn a meta concept of "baby-bottleness" and become able to recognize a baby bottle at any position and orientation, of any color and shape and so on....

The process continues as we grow up and we train more and more classifiers... but...

Baby are born as "tabula rasa". By being exposed to data produced by their senses they start understanding the world around them (©G.Longo).

Problems due to the lack of examples may arise also in grown-up people.

The classifier in our brain fails...

... if we turn the image upside down....

the classifier in our brain starts working ...

Why is it important to teach machines how to extract meaning from data?

Trivial answer

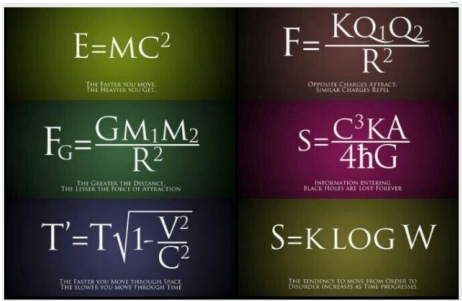
Because it empowers machine to perform tasks better and faster than humans in a huge variety of applications....

More Intriguing answer

Because it empowers humans to go beyond the limits of their own experiences and capabilities.

... Empirical laws are laws that can be confirmed directly by empirical observations. The term "observable" is often used for any phenomenon that can be directly observed, soit can be said that empirical laws are laws about observables... (R. Carnap, 1966).

But... almost all physical laws depend on max three independent variables....



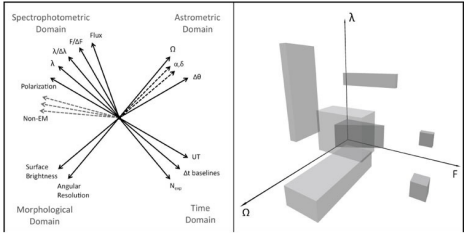
- Do we live in a simple universe?
- or it is just a bias induced by our brain in understanding the world?

Machines have no similar limits

SL and ML offer the possibility to find patterns, correlations, trends among many hundreds (or mmore) variables (parameters, features, ecc.). Thus allowing us to achieve a much more complex description of the world around us.

- Astrophysics
- Physics
- Chemistry
- Proteomics
- Climate studies
- Social sciences
- Finance
- Oceanography
- Medicine
- Agriculture
- ecc...

The Observed Parameter Space - OPS



An oversimplified representation of the OPS (©G.S. Djorgovski).

An important concept to which we shall return many times, is that of **Observable Parameter Space** or **OPS**. Records are composed by features, each feature adds a dimension to the OPS.

Each record with no NaN defines a point in the OPS. Records with 1 NaN define hyperlines in the OPS, with 2 NaNs define an hyperplane, and so on...

The problems we shall study are about finding complex patterns in these high dimensionality OPS.

	A	B	C	D	E	F	G
1	Customer ID	Last name	First name	Birthday	Country	Date of purchase	Amount of purchase
2		1 Bentham	Michael	06/03/1986	United States	08/12/2018	87
3		2 Vita	Jim	09/01/1994	United Kingdom	02/02/2018	80
4		3 Johnson	Fran	13/08/1973	France	10/11/2018	96
5		4 Lewis	Peter	20/10/1979	Germany	22/11/2018	27
6		5 Young	Richard	04/04/1970	United Kingdom	04/01/2018	67
7		6 Preston	Jack	16/04/1991	United States	06/11/2018	77
8		7 Smith	Daniel	13/03/1965	Canada	15/11/2018	32
9		8 Brown	Jack	03/09/1997	Australia	01/07/2018	75
10		9 Adline	Thomas	01/05/1986	Germany	07/11/2018	10
11		10 Williams	Ann	12/12/1980	United States	03/01/2018	18
12		11 Bennet	Alexis	12/04/1999	Canada	12/03/2017	56
13		12 Bond	James	25/02/1970	United Kingdom	12/08/2017	40
14		13 Bogdy	Patricia	23/12/1996	United States	20/01/2018	50
15		14 Roering	Michelle	02/04/1969	Canada	23/02/2017	29
16		15 Bailey	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
17							
18							

HOWEVER:
The dimensionality of the OPS can very easily explode....

A record of 8 different properties characterizing a customer defines a point in a 8-D space

A time series of 32000 samples (ex. a voice recording) can be considered ad a point in a 32.000-D space

The digital image of a painting can be considered as a point in a $4000 \times 4000 \times 3$-D space....

ecc...

OPS of this size are difficult wneh not plainly impossible to manage. This is were **Exploratory Data Analysis (EDA), feature engineering and selection, become crucial...**

Through a proper analysis, in fact, it is possible to reduce the number of dimensions (through rejection, combinations, ecc) without loosing or with a minimal loss of information.

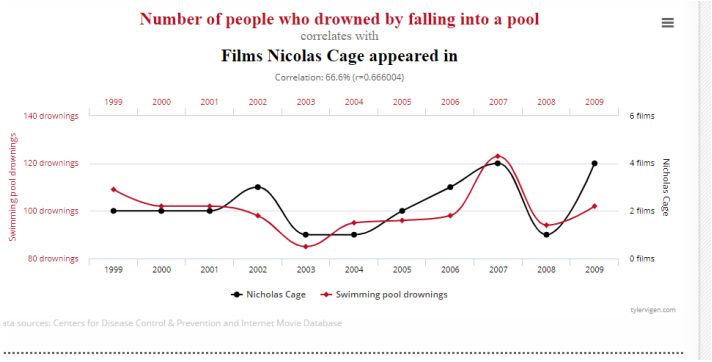
Patterns

A **pattern** is a structure or a trend in the OPS.

There is no univocous way to define a structure, and even less a significant structure. Usually they appear as correlations among parameters.

ML helps in finding such correlations also in spaces of high dimensionality.

The presence of a correlation does not mean that it is relevant.



Meaningfulness of a correlation

A correlation is useful if:

- it is easily understood by humans
- valid on new or test data with some degree of certainty
- potentially useful
- novel

Several objective measures of pattern interestingness do exist. For instance as we shall see in frequent pattern analysis, the so called **support** provides a good measure of significance for a pattern.

Features and Classes

A **data** is a set of instances. These instances are called **measures** (physics), **variables** (statistics), **parameters** (social sciences) or **features** (machine learning), ecc...

In the remaining we shall use the term FEATURE.

Data instances (features) can be associated with **classes** or concepts (e.g. class customer, class sales, class item, etc.).

The concept of class is truly useful only if "classes" can be specified in a unique way. The latter task is quite difficult and is the subject of extensive literature (under the keywords semantic and ontology).

Ontology

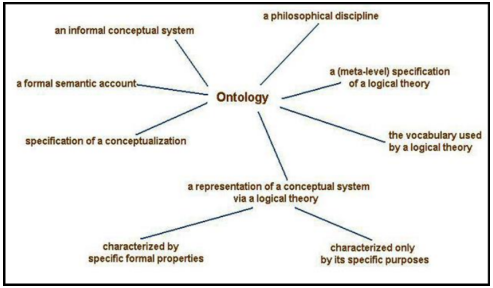


Figure 3: The seven definitions of onthology (©**Yadav et al. 2016**).

Ontology is a systematic account of Existence...

In information science appears in 1993, *... an ontology is the specification of a conceptualization....*

Any body of formally represented knowledge is based on a conceptualization (Logical Foundations of Artificial Intelligence, Genesereth & Nilsson, 1987).

A **conceptualization** is an abstract, simplified view of the world that we wish to represent for some purpose.

Every knowledge base, knowledge-based system, or knowledge-level agent is committed to some conceptualization, explicitly or implicitly.

... In the context of computer and information sciences, an ontology defines a set of representational primitives with which to model a domain of knowledge or discourse. The representational primitives are typically classes (or sets), attributes (or properties), and relationships (or relations among class members). The definitions of the representational primitives include information about their meaning and constraints on their logically consistent application. In the context of database systems, ontology can be viewed as a level of abstraction of data models, analogous to hierarchical and relational models, but intended for modeling knowledge about individuals, their attributes, and their relationships to other individuals. ... [...] in practice, the languages of ontologies are closer in expressive power to first-order logic than languages used to model databases. For this reason, ontologies are said to be at the semantic level, whereas database schema are models of data at the logical or physical level. Due to their independence from lower level data models, ontologies are used for integrating heterogeneous databases, enabling interoperability among disparate systems, and specifying interfaces to independent, knowledge-based services.

A short recap

If you are studying these slides you have very likely completed courses on Statistics and Statistical Data Analysis, hence you must be already acquainted with some basic concepts. However, given the different histories of the fields (statistical learning and machine learning) it is useful to summarize what some terms mean in the present course.

Statistical model:

is a set of mathematical functions that defines the behavior of the objects in a target class in terms of random variables and their associated probability distributions. In data mining statistical models are ubiquitous. They can either preeced the mining (i.e. data exploration and characterization, noise evaluation and removal, imputation of missing data, etc.) or be the outcome of the process.

Learning a task:

Machine learning does not understand a **"Task"** !! Learning in this context means that we learn how to perform the task.The distinction is subtle but it is crucial.
Example: As Goodfellow et al 2021 point how: *... If we want a robot to be able to walk, then walking is the task.* [and not learning how to walk... NdR], ... but...

... what does "learning" mean?

... Learning probabilities from data lies at the heart of machine learning and will be explored in great detail in this book... [Bishop & Bishop, 2024]

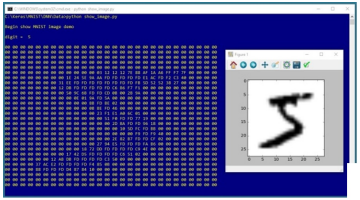
- Machine learning tasks are described in terms of how the machine learning system processes an example.
- An **example** data or *input* data is a collection of n **features** which have been estimated or quantitatively measured. Usually the data input are represented as **vectors**, $\mathbf{x} \in \mathbb{R}^n$, where x_i are the individual features.

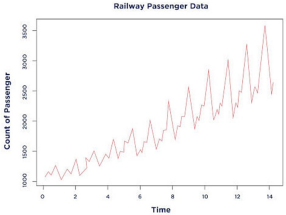
Examples:

- In image related tasks, the features are the values of the individual pixels of the images. Hence input data are matrices $D_{n,m}$ (if in greyscale), or datacubes ($D_{n,m,k}$).

- In a time series, the inputs are the values of one or many features measured at different epochs.

- ecc..





Tips & resources

TIPS

You wanto to become a data scientist (different from a data engineer). This means that you must be a scientist not an engineer.

1. Build your knowledge step by step in a critical way (better few things well understood than a lot of stuff downloaded from the internet but not digested)
2. Never use words or tools for which you do not know the detailed meaning or inner mechanisms.
3. Write codes for the future (your future). Respect programming rules, in ten years you want to be able to understand what you did.
4. If you download something from the internet write the source in the comments to your code. First for honesty (respect the work of others the same way you would like them to respect yours); Second for future use: in ten years from now you may want to reconstruct the reason why you wrote something.

Some references on semantics

For more information read these:

1. A Translation Approach to Portable Ontology Specifications<, T. R. Gruber, Knowledge Acquisition, 5(2):199-220, 1993
2. Tom Gruber's Humanistic AI pages. https://tomgruber.org/writing/definition-of-ontology
3. Semantic Web, see the W3C Data Activity web pages, https://www.w3.org/2013/data/

Resources

Resources on line. This lecture.

- **Data Management**:Picariello Lectures on Data Science: Roberto Maranca's crash course on Data Excellence.

Homeworks

First

Form the groups (max. 3 people, international, heterogeneous). Send the group's composition to me using the dedicated Teams channel.

- Find a common agreement on what "... learning..." means.
- Explore your group strengths and weaknesses (some of you can be better at math, other at programming, others have domain knowledge in other fields, ecc.) and summarize it in a file. Be honest. This will prove crucial later on.